

1 SFT

Instill Rigorous Reasoning

Supervised Fine-Tuning

Reverse-Perplexity Curriculum



Post-trained Reasoning Model
(Initialized from P1-30B-A3B)

Long-CoT SFT

Data Curation



Math



Code



Self-Verify



STEM



IF



Self-Correct

Data Curriculum

Unfamiliar

Proof-search Patterns



Familiar

High-PPL Examples



Low-PPL Examples



Post-Trained Reasoning Model
(Stronger Reasoning Style)

Goal: Install the rigorous reasoning pattern that the rest of the pipeline will later scale.

2 Coarse RL

Boost Reasoning Capabilities

Coarse Reinforcement Learning

Verifiable Rewards



Problem



Policy Rollouts

Policy



Rollouts

Outcome-Based Reward Model



Final Answer Correctness

$\int x dx$

Formal Verifier / Symbolic Check



Model-Based Judge



GSPO

Goal: Improve search, coverage, and direct solving performance on hard tasks.

3 Refined RL

Expand Reasoning Mechanisms

Refined Reinforcement Learning

Generative Rewards

1 Generation



Problem



Policy



Initial Solution

2 Self-Refinement



Initial Solution



Refine



Refined Solution

3 Experience Replay



Experience Buffer



Replay



Store



Policy



Policy Update



Generative Reward

DeepSeek-Math V2

✓ CoT Validity

✓ Proof Quality

✓ Anti-Hacking

Goal: Use more fine-grained feedback to encourage proof rigor and self-refinement.

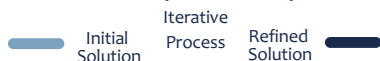
4 TTS

Scale Test-Time Reasoning

Model-Agnostic Test-time Scaling

Verification-and-Refinement Loop

Iterative Self-Improvement Pipeline



Review ↓ Feedback ↑

Bug Report

● Critical Errors

● Major Justification Gaps

● Minor Justification Gaps

Accept

Pass Verification
5 Consecutive Times

Reject

Major Issues
Persist for 10 Steps

Goal: Scale test-time compute via iterative refinement to enhance answer quality.



Model / Policy



Input / Problem



Output / Answer



Experience Buffer



Iterative Process



Optimization